# Xuanzhi Chen

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## Research Interests

Spatial multi-omics, machine learning for biology, and scientific agents.

## Education

**Peking University** Beijing, China
*B.Eng. Candidate in Bioinformatics, School of Life Sciences* *Sep. 2023 – Jun. 2027 (Expected)*

- **GPA: 3.77/4.0.** Pursuing a double degree in Economics at the National School of Development.
- **Selected coursework:** Computational Immuno-oncology (99/100); Data Structures and Algorithms (98/100); Statistical Analysis of Genomics Data (A); AI Methods and Applications in Life Sciences (A).

## Preprint

Nianping Liu, **Xuanzhi Chen**, Miao Cui, Xiaoying Liao, Xiaoke Song, Sairam Pantham, Weize Xu, and Xiaojie Qiu. **Bio-Babel: autonomous cross-language reconstruction of agent-ready computational biology ecosystems.** *bioRxiv*, 2026. https://doi.org/10.64898/2026.08.30.743167.

## Research Experience

**College of Future Technology, Peking University** Beijing, China
*Undergraduate Researcher; Advisor: Prof. Hu Zeng* *Jul. 2024 – Present*

- **STATES-GBM:** Built and optimized upstream processing for STATES, which jointly profiles spatial transcriptomes and translatomes. Benchmarked cell-segmentation methods, integrated 3D Cellpose nuclear segmentation into ClusterMap, and improved image stitching. Reduced seams misaligned by $> 3$ pixels from 14.2% to 0.3% and multi-nucleus segments from 28.8% to 13–14% against the same nuclear reference. Analyzed malignant-cell identities, subtypes, and spatial variation in translation efficiency in glioblastoma.
- **iST Agent:** Built an agent for reliable execution of heterogeneous imaging-based spatial transcriptomics workflows, separating source-derived contracts from memory of runtime failures and fixes. Generated contracts for seven pipelines and reproduced one pipeline's decoding byte for byte. In one comparison, preflight checks caught a multi-sample configuration defect with memory but missed it without memory.

**Department of Genetics, Stanford University** Stanford, CA
*Undergraduate Researcher; Advisor: Prof. Xiaojie Qiu* *Jun. 2026 – Present*

On-site summer research, Jun.–Aug. 2026; continued remotely.

- **Bio-Babel:** Built an annotation agent that generates machine-readable usage contracts for reconstructed packages, enabling downstream agents to access their functionality. Extended coding-agent compatibility and developed tooling for agent-driven Python-to-C++ reconstruction. Conducted the pancreatic differentiation case study, including preprocessing and cell annotation, followed by analysis through Bio-Babel's agents. Reviewed trajectory analyses and completed genotype-aware screening and biological interpretation of reduced maturation signatures under SWI/SNF loss.
  https://github.com/Bio-Babel/UMI-tools-cpp | https://github.com/Bio-Babel/bio-babel-pancreas
- **End-to-end spatial transcriptomics modeling:** Developing and training a model on public MERFISH data that couples molecular localization and barcode decoding with cell segmentation and molecule-to-cell assignment. The architecture combines RNA fluorescence and cellular morphology, learning cross-round correspondences without a separate image-warping stage. The project tests whether sharing evidence across these interdependent tasks can reduce errors introduced by sequential processing.

**Canchen Technology** Remote
*Research Intern; Mentor: Dr. Lei Li* *Oct. 2026 – Present*

- **Agentic virtual tissue modeling:** Investigating hybrid world models that integrate foundation models with mechanistic simulations, connecting learned cell-state representations with explicit biological dynamics.
- **Agent-guided prediction:** Designing bidirectional feedback between foundation models and LLM agents, with trainable interfaces for agents to modulate features, select model components, and iteratively refine predictions using literature and biological context.

## Selected Presentations

**Building an Agentic Tissue World Model.**
Journal club presentation, Canchen Technology, Nov. 2026.

**Bio-Babel: autonomous cross-language reconstruction of agent-ready computational biology ecosystems.**
Invited oral presentation, PKU Undergraduate Honors Program in Biology Annual Symposium, Oct. 2026.

**Bio-Babel and End-to-End Spatial Transcriptomics Modeling.**
Research presentation, Qiu Lab, Stanford University, Aug. 21, 2026.

**How to Build the Virtual Cell?**
Group course presentation, Systems Biology, Peking University, Jun. 6, 2026.

**SpatialFusion: A Lightweight Multimodal Foundation Model for Pathway-Informed Spatial Niche Mapping.**
Journal club presentation, Zeng Lab, Peking University, May 27, 2026.

## Honors and Awards

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| Academic Excellence Award, Peking University | 2026 |
| Peking University Scholarship | 2025 |
| Peking University Freshmen Scholarship | 2023 |
| Gold Medal, China National Biology Olympiad Final | 2022 |

## Leadership & Activities

**Volleyball Enthusiasts Association, Peking University**

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| Chair | 2024–2025 |
| Director of Competitions | 2023–2024 |

**School of Life Sciences Volleyball Team, Peking University**
Captain

## Technical Skills

**Experimental:** In situ spatial omics (STATES), molecular cloning, CRISPR-Cas9, PCR/qPCR, Western blotting.

**Computational:** Python, R; single-cell and spatial omics, image analysis, scientific agents, workflow automation.